Physics School
Engineering Physics Licentiate Degree
Final Graduation Project

Analysis of magnetohydrodynamics oscillations at Heliotron J with Python

Student:
Kevin Enrique Rishor Villalobos
2022093578
krishor@estudiantec.cr

April, 2026

Heliotron J Project, Institute of Advanced Energy
Kyoto University

1 Introduction

1.1 Identification

1.1.1 Institution

Located at Gokasho, Uji, Kyoto 611-0011, Japan, the Heliotron J project is a research initiative of the Institute of Advanced Energy at Kyoto University, which focuses on the advancement of controlled fusion. This experimental facility utilizes a helical-axis heliotron configuration to investigate the magnetic confinement and contribute to understanding how to achieve the stability of high-temperature plasmas. By optimizing magnetic field geometry and advanced heating methods, the project aims to establish the scientific and engineering foundations required for the development of a sustainable, fusion-based power generation system.

1.1.2 Contact Information

Name	Ph.D. Prof. Shigeru Inagaki
Email	`inagaki.shigeru.7s@kyoto-u.ac.jp`
Tel	+81-774-38-3484

1.2 Justification

Nuclear fusion is widely recognized as a promising solution to future global energy demands; however, its realization critically depends on the sustained confinement of high-temperature, high-density plasmas. A major obstacle to this goal is the presence of magnetohydrodynamic (MHD) instabilities, which can severely degrade plasma performance. In helical devices such as Heliotron J, MHD oscillations driven by energetic electrons and/or ions—particularly Alfvén eigenmodes—lead to significant transport and loss of high-energy particles. These processes not only reduce heating efficiency and energy confinement but also pose a serious risk of localized damage to plasma-facing components through the formation of particle hot spots [1].

Recent experimental observations in Heliotron J further reveal that low-frequency electromagnetic fluctuations are excited during the H-mode phase. These fluctuations couple strongly with edge turbulence, enhancing cross-field heat transport and thereby limiting the expected improvement in energy confinement [2]. Such behavior is considered a key factor underlying the relatively modest H-mode performance in helical devices compared to tokamaks. In addition, the excitation of large-amplitude MHD oscillations has been observed under certain operational conditions, highlighting the complexity and variability of plasma stability in these systems [3]. These findings underscore the urgent need for a comprehensive understanding of both the characteristics and driving mechanisms of MHD oscillations in order to achieve stable plasma operation.

This thesis graduation project aims to bridge the gap between present experimental knowledge and the requirements of a practical fusion reactor by systematically investigating MHD oscillations in Heliotron J. The focus is placed on energetic particle-driven modes [1] and low-frequency fluctuations arising during H-mode [2], with the goal of identifying the physical mechanisms through which these instabilities limit energy confinement.

To this end, we employ multi-channel magnetic fluctuation diagnostics to resolve the spatiotemporal structure of MHD activity. Advanced time-series analysis and higher-order correlation techniques are applied, including Fourier, Hilbert, and wavelet transforms, to extract coherent structures and causal relationships from complex plasma signals [2]. By visualizing these correlations, we aim to clarify how specific plasma conditions and magnetic configurations trigger MHD oscillations [3].

Beyond understanding, this study also targets the control of MHD instabilities. Heliotron J offers a uniquely flexible magnetic configuration, enabling systematic exploration of plasma behavior across a wide range of magnetic well depths and rotational transform profiles [4]. Identifying configurations that suppress MHD activity will provide a direct pathway to improved confinement performance.

Ultimately, understanding and controlling MHD instabilities is essential for next-generation fusion devices such as ITER and DEMO, where achieving net energy gain is imperative. By uncovering the driving mechanisms of these oscillations in a flexible, well-diagnosed experimental platform, this research will contribute to the development of predictive capabilities and control strategies that are indispensable for the realization of practical fusion power plant [5].

1.2.1 Hypothesis

It is theorized that the MHD oscillations observed in Heliotron J are determined by specific plasma conditions and magnetic configurations, and that their systematic analysis through Python-based signal processing will identify the dominant excitation mechanisms responsible for these oscillations. Furthermore, it is expected that energetic-particle-driven modes and low-frequency H-mode fluctuations can be characterized by distinct spectral and temporal signatures, which will correlate with plasma heating performance, transport behavior, and confinement degradation or improvement under different operational conditions.

1.3 Objectives

1.3.1 General Objective

To characterize MHD oscillations in the Heliotron J device through Python-based analysis of experimental data, with the purpose of determining their excitation mechanisms and assessing their impact on plasma confinement.

1.3.2 Specific Objectives

- To characterize the spectral properties and temporal evolution of MHD oscillations in experimental discharges through Python-based signal processing techniques, with the generation of time-frequency representations and the identification of dominant modes.

- To evaluate the correlation between energetic particle modes and plasma heating performance through the analysis of synchronization among magnetic fluctuation frequencies, Neutral Beam Injection (NBI), and Electron Cyclotron Heating (ECH), with the derivation of statistical correlation metrics.
- To identify low-frequency H-mode fluctuations through the analysis of magnetic signals, with the construction of a detailed mapping of nonlinear coupling events associated with edge turbulence modulation and enhanced heat transport.

2 Preliminar Methodology

2.1 Preliminar Procedure

2.2 First Specific Objective

For the first specific objective, the study will begin with a literature review of relevant and related works on the Heliotron J device. Subsequently, data analysis will be carried out using Python libraries developed and released by the Heliotron J team. In this stage, signal processing scripts employing techniques such as FFT and wavelet analysis will be examined, refined, and further developed. These scripts will then be applied to magnetic probe data obtained from past experiments, and through visualization of correlations, the dominant oscillation modes present in the plasma will be identified.

The FFT analysis will be performed with segment lengths with a frequency resolution of $\Delta f = \frac{1}{T}$, where T is the window duration. For Alfvén eigenmode detection, these lengths will be selected to balance frequency resolution against temporal stationarity of the signal, this for consistency with the sampling rates of the Mirnov coils. Also, to reduce the spectral leakage in the FFT, a Hanning window function will be applied to each segment. For the wavelet analysis, a Morlet mother wavelet will be employed, which is suitable for the non-stationary character of MHD oscillations in plasma discharges. A mode will be considered dominant if its spectral power exceeds a signal-to-noise threshold of at least 3σ above the background noise level. Additionally, spatial coherence between multiple Mirnov coil channels, $\gamma^2 > 0,7$, will be required to confirm that the identified feature corresponds to a physically coherent MHD structure rather than just instrumental noise.

The signal processing pipeline will rely primarily on the open source Python libraries developed and released by the Heliotron J team, available at the Heliotron J Project webpage. These include *libana_signal* for core signal analysis routines, *anahilbert* for Hilbert transform-based envelope and instantaneous frequency extraction, *anaaspect* for spectral analysis, *analpf* for low-pass filtering, *jgraph* for data visualization, and *jview* for data browsing. These will be complemented by standard scientific Python libraries such as NumPy and SciPy for FFT computation and array operations, and Matplotlib for figure generation.

2.3 Second Specific Objective

The analysis will focus on Neutral Beam Injection experiments. Frequency variations of magnetic fluctuations, which are considered to be associated with energetic particle losses, will be evaluated and compared with models of energetic particle distribution function variations. This will enable validation of the interactions between energetic particles and waves.

The correlation between magnetic fluctuation frequencies and the NBI/ECH heating signals will be evaluated using cross-correlation analysis in the time domain and cross-spectral coherence in the frequency domain. The instantaneous frequency of identified magnetic modes will be extracted via the Hilbert transform, using the *anahilbert* package, and subsequently, compared against the NBI injection timing and power waveforms. The degree of association between fluctuation amplitude/frequency and heating power will be quantified with the Pearson correlation coefficient r, computed across different experimental discharges under varying NBI power levels. A correlation will be considered statistically significant if $|r| > 0,7$ with a p-value $<$ 0,05.

Finally, the results obtained in this section will be validated by comparing the measured mode frequencies against the theoretical predictions for energetic-particle-driven modes, specifically, by checking consistency with the Alfvén continuum and the expected frequency scaling with the Alfvén velocity derived from equilibrium plasma parameters. If agreement between measured frequency sweeping behavior and energetic particle distribution function models is reached, this will serve as the primary validation criterion, following the approach of Zhong et al. [1].

2.4 Third Specific Objective

For the final objective, data analysis of low-frequency EM fluctuations during H-mode will be conducted. This analysis will also include electrostatic fluctuations, and by examining their nonlinear coupling, a deeper understanding of edge turbulence modulation by electromagnetic fluctuations will be obtained. Finally, the results and discussions will be compiled into a final report, which will be thoroughly reviewed and refined prior to submission.

A low-frequency electromagnetic fluctuation will be identified as a candidate modulating mechanism if it satisfies the three-wave nonlinear coupling condition $f_1 \pm f_2 = f_3$, verified through bicoherence analysis. This higher-order spectral technique, consistent with the approach of Kin et al. [2], allows distinction between linearly excited modes and nonlinearly coupled ones. A bicoherence value $b^2 > 0,5$ will be adopted as the threshold for identifying statistically significant three-wave interactions between electromagnetic fluctuations and edge turbulence. The identified coupling events will be validated through two complementary approaches:

- **Cross-diagnostic consistency:** Verifying that the same coupling signatures appear simultaneously in both the magnetic (Mirnov coils) and electrostatic (Langmuir probes) channels.
- **Discharge reproducibility:** Confirming that the identified nonlinear coupling patterns persis across multiple H-mode discharges under similar operational conditions.

Agreement between the enhanced heat transport periods and the temporal windows where strong bicoherence is detected will serve as the final confirmation that the identified mechanism is physically relevant to confinement degradation.

2.5 Preliminar Project Schedule

Table 1: Preliminary schedule, grouped by activities and specific objectives

Specific Objective	**Activity**	**Weeks**															
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
To characterize the spectral properties and temporal evolution of MHD oscillations using Python-based signal processing to identify dominant modes during experimental discharges.	Literature review and Python environment setup																
	Development and refinement of signal processing scripts (FFT and Wavelets)																
	Extraction of magnetic probe signals and correlation visualization																
	Identification and classification of dominant oscillation modes																
To evaluate the correlation between energetic particle-driven modes and plasma heating performance (NBI/ECH) to clarify their impact on energetic particle transport.	Frequency variation analysis of magnetic fluctuations during NBI/ECH																
	Comparison with energetic particle distribution models and wave-particle validation																
	Assessment of impact on transport and heating efficiency																
To identify low-frequency fluctuations excited during H-mode and elucidate the mechanisms underlying edge turbulence modulation and enhanced heat transport.	Analysis of low-frequency EM and electrostatic fluctuations in H-mode																
	Investigation of nonlinear coupling and edge turbulence modulation																
	Writing of the final report, results, and conclusions																
	Final document review and defense preparation																

3 Use of resources

3.1 Technical, material and physical aspects

3.1.1 Technical

Ph.D. Prof. Shigeru Inagaki, and Ph.D. Prof. Kazunobu Nagasaki, allong with other collaborators at the Heliotron-J Project will provide me the academic mentorship needed to support my research. Also, Ph.D. Prof. Iván Vargas-Blanco will be my thesis supervisor, providing his knowledge and research experience in Stellarator devices. Finally, M.Sc. Prof. Ricardo Solano-Piedra will be my thesis lector, helping reviewing and affining my research.

3.1.2 Material

- Computing Hardware
  - Personal computer for large-scale signal processing.
  - Data storage units and SSDs for the management of high-frequency experimental datasets.
  - Local servers and central data acquisition systems located at the Institute of Advanced Energy.

- Heliotron J Experimental Device
  - The L/M = 1/4 helical-axis heliotron stellarator device.
  - Magnetic field coil systems including helical, toroidal, and vertical coils for magnetic configuration control.
  - Vacuum vessel and structural components that maintain the plasma environment.
  - Neutral Beam Injection (NBI) heating systems used as the source for energetic-particle-driven oscillations.
  - Electron Cyclotron Heating (ECH) and Current Drive (ECCD) gyrotron systems and transmission lines.
- Diagnostic Instrumentation and Sensors
  - Mirnov coil arrays consisting of multiple magnetic probes for the detection of magnetic field fluctuations.
  - Soft X-ray (SX) detector cameras and silicon photodiode arrays for tomographic measurement of internal oscillations.
  - Electron Cyclotron Emission (ECE) radiometers and antennas for measuring electron temperature fluctuations.
  - Beam Emission Spectroscopy (BES) hardware for the observation of density fluctuations in the plasma core.
  - Microwave and laser interferometers for the measurement of line-averaged electron density.
  - Langmuir probe assemblies for the physical measurement of edge plasma turbulence and electric fields.

3.1.3 Physical

As stated before, the project will take place physically at the Heliotron J Project Laboratory, at the Institute of Advanced Energy in Kyoto University. Located in Gokasho, Uji, Kyoto 611-0011, Japan

3.2 Budget

The technical equipment and data access are provided by the host institution or already possessed by the student, as shown in Table 2. The operational and living expenses, which will be covered by the student, are detailed in Table 3.

Table 2: Budget for materials and tools

Description	Quantity	Unit Value	Subtotal	Available	Comments
Personal computer	1	600,00	600,00	Yes	Linux-based laptop
Software licenses	1	0,00	0,00	Yes	Open-source Python libraries
Heliotron J device	1	0,00	0,00	Yes	Provided by Kyoto University
Diagnostic data	1	0,00	0,00	Yes	Internal database access
Total			600.00		

Table 3: Budget for services (3-month stay in USD)

Description	Value	Comments and observations
Airfare	2430,00	Round-trip international flight
Housing	497,33	Uji House (includes admission fee and rent)
Food	570,00	Budget based on home-cooking
Academic fees	1158,00	Admission and monthly tuition fees
International medical insurance	150,00	Required coverage for the full stay
Mobile data plan	90,00	High-speed internet for mobile device
Utilities	105,00	Estimated electricity and gas consumption
Internal transportation	120,00	Bycicle insurance, and helmet
midrule Total	5120,33	Total project execution cost

Note: Calculations are based on an exchange rate of 1 USD = 150 JPY.

References

[1] Y. Zhong, K. Nagasaki, Z. Wang, S. Kobayashi, S. Inagaki *et al.*, "Response of energetic-particle-driven magnetohydrodynamic (mhd) instability to modulated ech in heliotron j," *Plasma and Fusion Research*, vol. 19, p. 1202008, 2024.

[2] F. Kin, Y. Kondo, M. Sasaki, A. Miyashita, R. Matsutani, K. Ozawa *et al.*, "Investigation of low-frequency fluctuation modulating turbulence during l-h transition in heliotron j," *Physics of Plasmas*, vol. 32, no. 7, p. 072510, 2025.

[3] A. Miyashita, S. Ohshima, Z. Xu, F. Kin, S. Kobayashi, S. Kado *et al.*, "Transition from divertor to limiter configuration and associated transient plasma oscillation in heliotron j," *Plasma Physics and Controlled Fusion*, vol. 67, no. 4, p. 045003, 2025.

[4] F. Sano *et al.*, "Physics of heliotron j confinement," *Plasma and Fusion Research*, vol. 5, p. S2003, 2010.

[5] K. Nagasaki, S. Yamamoto, S. Kobayashi, K. Sakamoto, Y. I. Nakamura, Y. Nagae *et al.*, "Stabilization of energetic-ion-driven MHD mode by ECCD in Heliotron J," in

Proceedings of the 24th IAEA Fusion Energy Conference (FEC 2012), ser. EX/P8-10. San Diego, USA: IAEA, 2013.